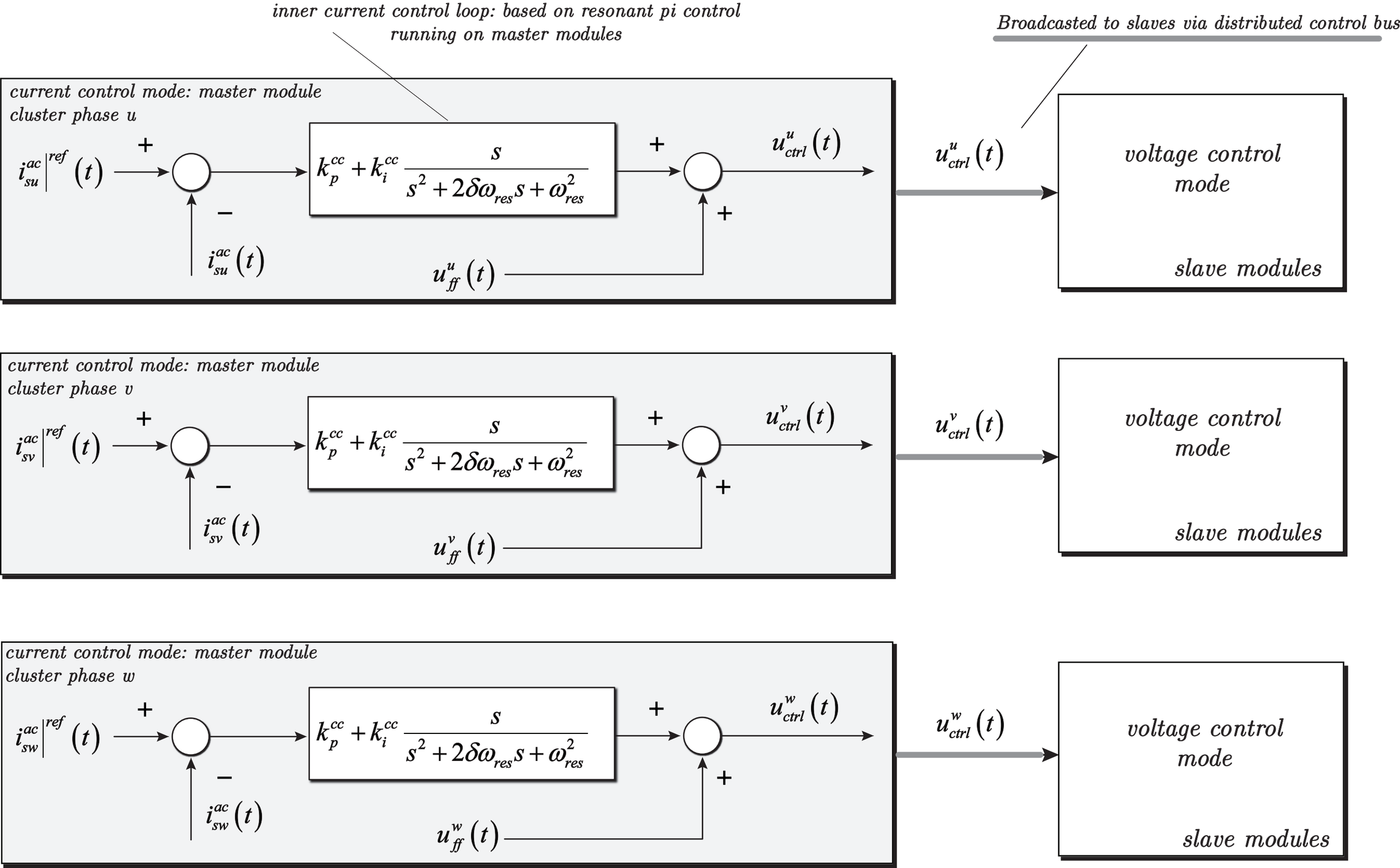

inner current control loop: based on resonant pi control
running on master modules
Broadcasted to slaves via distributed control bus
current control mode: master module
cluster phase u
$i_{su}^{ac}\big|^{ref}(t)$
+
−
$k_p^{cc} + k_i^{cc}\frac{s}{s^2 + 2\delta\omega_{res}s + \omega_{res}^2}$
+
+
$u_{ctrl}^{u}(t)$
$i_{su}^{ac}(t)$
$u_{ff}^{u}(t)$
$u_{ctrl}^{u}(t)$
voltage control mode
slave modules
current control mode: master module
cluster phase v
$i_{sv}^{ac}\big|^{ref}(t)$
+
−
$k_p^{cc} + k_i^{cc}\frac{s}{s^2 + 2\delta\omega_{res}s + \omega_{res}^2}$
+
+
$u_{ctrl}^{v}(t)$
$i_{sv}^{ac}(t)$
$u_{ff}^{v}(t)$
$u_{ctrl}^{v}(t)$
voltage control mode
slave modules
current control mode: master module
cluster phase w
$i_{sw}^{ac}\big|^{ref}(t)$
+
−
$k_p^{cc} + k_i^{cc}\frac{s}{s^2 + 2\delta\omega_{res}s + \omega_{res}^2}$
+
+
$u_{ctrl}^{w}(t)$
$i_{sw}^{ac}(t)$
$u_{ff}^{w}(t)$
$u_{ctrl}^{w}(t)$
voltage control mode
slave modules